AES Senior Project RFP – Section 013 and 014

1.0 Project Title:

An Autonomous and Cooperative UAS Multirotor and Rover Network for Exploration, Search, and Payload Delivery/Retrieval

2.0 PI:

Melvin Rafi

3.0 Project Background:

Give a brief summary of the field of application and relevant previous work. Suggest any resources students should review.

In search and exploration missions, it may sometimes be advantageous to utilize a multi-agent network of air and land-based vehicles to explore an area of interest and to deliver or retrieve payloads at locations with reduced accessibility, where the precise location may not be initially known.

Within this multi-agent network, a UAS multirotor may take on the roles of search, exploration, and determination of a precise location of interest (for example, a stranded hiker), while a rover may take on the roles of payload delivery or retrieval once the precise location has been determined by the UAS. While the UAS has the advantage of maneuvering over terrain to perform a wider area search, it may not have the ability to land in areas lacking sufficient clearance. In such a case, the rover has the advantage of maneuvering on terrain inaccessible by the UAS to deliver or retrieve a payload at the desired location.

In this context, each vehicle autonomously and cooperatively works together to increase the overall chances of mission success.

4.0 Mission Objective:

Outline the desired mission this design must perform to be successful.
Faculty information: students will utilize this to iterate or develop their CONOPS. Mission Objectives (MOs) are defined as the broad goals which the system must achieve to be productive.
Example MO statement from SMAD for a satellite which monitors forest fires: 'To detect, identify, and monitor forest fires through the United States including Alaska and Hawaii in near real time.'

The objective of the UAS multirotor and rover network is to autonomously and cooperatively explore an area of interest to locate and deliver/retrieve a payload at a desired location of interest.

5.0 Project Description:

Summarize the design problem.
Faculty information: Describe the problem you want to the students to solve. Outline goals of what a successful system must accomplish. Motivate how solving this problem will impact the field.

This project involves developing and prototyping a network comprising of one UAS and one rover that can function autonomously (or semi-autonomously) and cooperatively. A successful vehicle network

should be able to explore an area of interest, determine a precise location for payload delivery/retrieval, and deliver/retrieve a payload at the desired location.

6.0 Initial Requirements:

The following capabilities/attributes are considered the initial requirements which must be met for the prototype to be considered successful; however, final requirements may differ based on project team analysis and trade studies. All requirements provided below are expected to be questioned and evaluated for necessity, appropriateness for mission and feasibility. Any changes, refinements or additions to these requirements must be coordinated with the PI (faculty advisor) for this program.

Faculty: please list a minimum of 5 and maximum of 10 FRs. FRs are defined as 'How well the system must perform to meet its mission objectives.'

Examples of previous FRs:

1) *Aircraft must be human-portable at full mission capability. Single operator must be able to carry a minimum of 2x air vehicles (and any required launch/recovery equipment)*
2) *The spacecraft shall have daily coverage of 750 million acres within continental US*
3) *The spacecraft shall be able to detect a fire with a 500 m locational accuracy*

General Requirements

1) The multi-agent network should comprise one UAS and one rover working autonomously/semi-autonomously and cooperatively to search an area of interest, determine the coordinates of a location of interest, and deliver/retrieve a payload at the location of interest, which is specified by an ArUco marker.

UAS Team

1) The UAS multirotor platform should be able to contain all necessary hardware, sensors, communications equipment, and miscellaneous electronics, as required, to support safe fulfilment of the mission objectives.

2) The UAS shall have a volumetric footprint of at most 50 cm length x 50 cm width x 30 cm height, shall have an endurance of at least 15 minutes, shall be able to function and operate in temperatures ranging from 15°C to 25°C, and shall weigh no more than 25 lb (11.3 kg).

3) The UAS shall be able to takeoff from a starting location, autonomously maneuver to a predetermined area of interest, autonomously search the area of interest to detect an ArUco marker to within a 30 cm radius of accuracy, relay the coordinates of the marker to the rover or a base station, and return to its starting location.

4) For testing and demonstration of the prototype, the UAS shall maneuver at an altitude no higher than 2m above ground level, at a speed of no more than 0.5 m/s, and must remain within visual line of sight (the team should ensure that these parameters are in compliance with and approved by CU Flight Ops).

5) The autonomous phases of flight shall be carried out semi-autonomously where necessary or appropriate. Each autonomous phase of flight must have an override for manual control.

6) The UAS must comply with all relevant FAA regulations and CU Flight Ops policies.

Rover Team

1) The rover platform should be able to contain all necessary hardware, sensors, communications equipment, and miscellaneous electronics, as required, to support safe fulfilment of the mission objectives.

2) The rover shall have a volumetric footprint of at most 30 cm length x 30 cm width x 30 cm height (excluding the payload delivery/retrieval mechanism), shall have an endurance of at least 60 minutes, and shall be able to function and operate in temperatures ranging from 15°C to 25°C.

3) The rover shall be able to receive coordinates of the ArUco marker from the UAS or a base station, autonomously move from a starting location to the marker location, deliver/retrieve a payload within a 30 cm radius of the marker location, and return to its starting location. Payload delivery/retrieval may be conducted via manual control of the rover.

4) The rover shall be able to navigate over terrain features of 3cm height without having its movement impeded.

5) The autonomous phases of motion shall be carried out semi-autonomously where necessary or appropriate. Each autonomous phase of motion must have an override for manual control.

Definitions

1) 'Area of interest' is defined for testing purposes as a space of approximately 5m x 5m. Testing may be conducted indoors or outdoors, depending on availability of labs or outdoor spaces. All testing must meet the necessary safety requirements and approvals before testing can commence.

2) 'Location of interest' is defined as a singular coordinate where an ArUco marker is located.

3) 'Payload' is defined as an object of size 8cm x 8cm x 8cm and mass of no more than 1kg.

4) 'Terrain feature' is defined as an object of at least 3 cm height.

Additional Notes

1) Section 013 will focus on the payload delivery requirement for the rover, while Section 014 will focus on the payload retrieval requirement for the rover.